(a) Subject-disjoint data protocol

Training: six subjects

81 ⊂ 162 ⊂ 324 clips

Validation: two subjects

108 clips; checkpoint + L2

Test: two subjects

108 clips; selected head only

Train: 1, 3, 4, 6, 7, 8 | Validation: 5, 10 | Test: 2, 9

(b) Frozen-backbone transfer pipeline

Video input

5 fps; 172² RGB

→

MoViNet A0

Frozen backbone

→

Clip features

480 dimensions

→

Standardize

Training statistics

→

Trainable head

Dropout → Dense

→

Prediction

18 actions

Train-only normalization • 8,658 trainable parameters • Validation-only model selection

Fresh backbone states per clip; no video augmentation or person crops.